# SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

## Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

### Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 $m^2$

### Renovation Details by Scenario

| Scenario | Renovation Type | Renovation Details |
|---|---|---|
| Baseline | baseline | Baseline (no envelope renovation) |
| Scenario 1 | wall + roof + window + door | **Wall upgrade:**<br>• Exterior wall insulation: Mineral Wool Heavy Density Blanket, target R-30.0<br>**Roof upgrade:**<br>• Roof insulation: Graphite Polystyrene (GPS) Foam Board, target R-28.0 |

**Window upgrade:**

- 2-pane replacement (skipped: SimpleGlazing windows)
- Weatherstrip application (skipped: no OperableWindow)
- Glazing film application (skipped: SimpleGlazing windows)
- Caulking application: polyurethane
- Window frame replacement (skipped)

**Door upgrade:**

- Door replacement: honeycomb core steel door (1 of 3 door(s) skipped: door type incompatibility)
- Bottom seal: brush weatherstrip
- Top/side seal: jamb weatherstrip

## Annual Energy Analysis

| Scenario | Metric | Baseline | Renovation | Delta | Savings % |
|---|---|---|---|---|---|
| Scenario 1 | Total Site Energy (GJ) | 236.54 | 223.23 | **13.31** | **5.63%** |

## Key Performance Metrics

### Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline $9,841

Scenario 1 $9,383

Baseline Scenario 1

Max saving: **$-459 (-4.66%)**

### Operational Carbon Saving (Scenario - Baseline) (kg $CO_2e$/yr) Comparison

Baseline 20,894

Scenario 1 20,001

Baseline Scenario 1

Max saving: **-893 kg $CO_2e$ (-4.27%)**

Min Retrofit Construction Cost

**$71,867**

Scenario 1

Min Retrofit Embodied Carbon

**9,156 kg $CO_2e$**

Embodied carbon intensity (ECI): 18 kg$CO_2$e/m$^2$

Scenario 1

Lowest Retrofit Cost Payback Period

**157 years**

Scenario 1

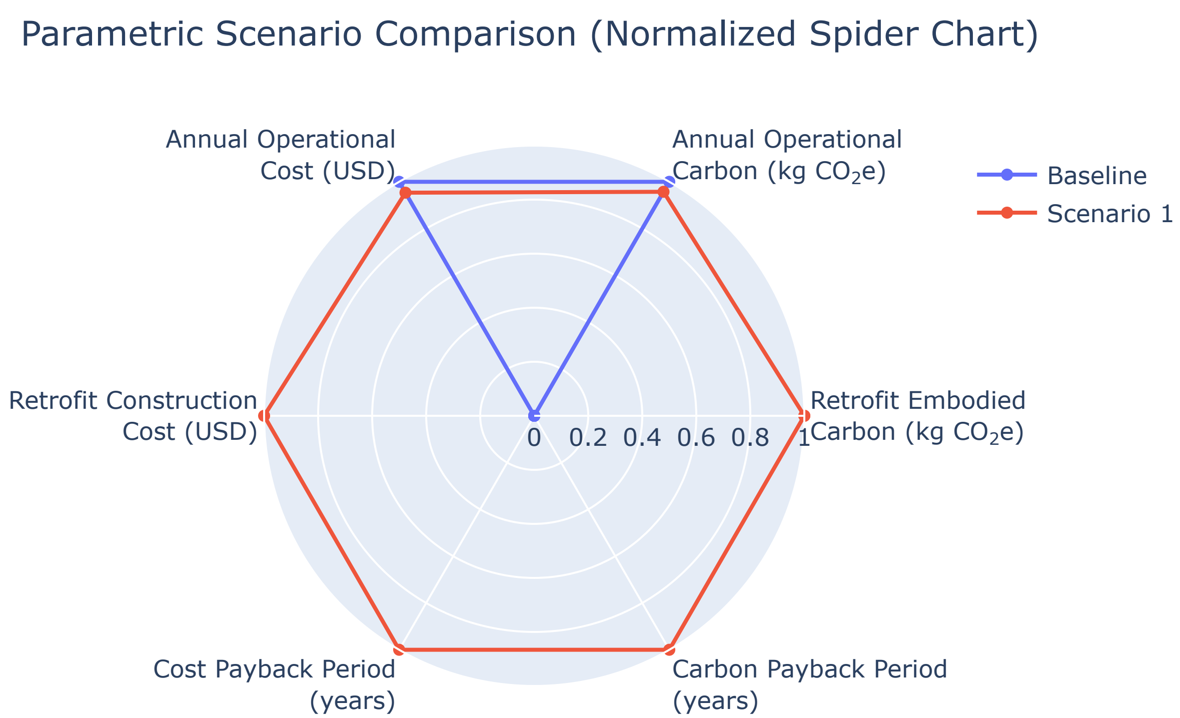
Lowest Retrofit Carbon Payback Period

**10 years**

Scenario 1

## Comparative Visualizations between Renovation Scenarios

### Spider Chart Visualization




## Payback Period of the Retrofit

### Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 157 yrs

### Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg $CO_2$e/yr)).

Scenario 1 10 yrs

## Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

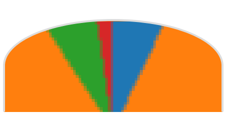
| Scenario | Component | Material name | Volume ($m^3$) | Area ($m^2$) | Thickness (m) | Length (m) | Thermal Conductivity (W/m·K) | Density ($kg/m^3$) | Product Lifetime (years) |
|---|---|---|---|---|---|---|---|---|---|
| Scenario 1 | Wall insulation | Mineral Wool Heavy Density Blanket | 20 | 282 | N/A | N/A | 0.03 | 103 | 30 |
| Scenario 1 | Roof insulation | Graphite Polystyrene (GPS) Foam Board | 116 | 599 | 0.19 | N/A | 0.03 | 14 | 75 |
| Scenario 1 | Window caulking | polyurethane | 0.0034 | N/A | N/A | N/A | N/A | N/A | 10 |
| Scenario 1 | Door | honeycomb core steel door | N/A | 3.90 | N/A | N/A | 0.05 | 482 | 30 |
| Scenario 1 | Door bottom sealing | brush weatherstrip | N/A | N/A | N/A | 3.66 | N/A | N/A | 15 |
| Scenario 1 | Door top/side sealing | jamb weatherstrip | N/A | N/A | N/A | 16 | N/A | N/A | 15 |

## Result Summary

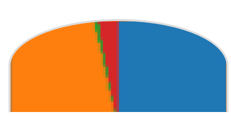
Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

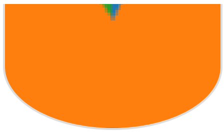
### Scenario 1

#### Cost breakdown by retrofit measure



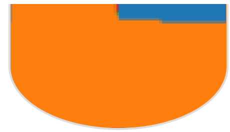
#### Carbon breakdown by retrofit measure





- Wall: $5,100 (7.1%)
- Roof: $60,118 (83.7%)
- Window: $4,953 (6.9%)
- Door: $1,695 (2.4%)

Total: $71,867



- Wall: 2,334 (25.5%)
- Roof: 6,499 (71.0%)
- Window: 52 (0.6%)
- Door: 270 (3.0%)

Total: 9,156 kg $CO_2e$

| Scenario | Retrofit Embodied Carbon (kg $CO_2e$) | Retrofit Construction Cost (USD) | Annual Operational Carbon (kg $CO_2e$) | Annual Operational Cost (USD) |
|---|---|---|---|---|
| Baseline | 0.00 | $0.00 | 20,894 | $9,841 |
| Scenario 1 | 9,156 | $71,867 | 20,001 | $9,383 |

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